Worked example: Linux syslog into the entity index

No parsing at the collector. Every step below is plain KQL that runs in the lake.

RAW LINES AS THEY LAND: PROCESSNAME AND SYSLOGMESSAGE ONLY

Collector adds no parsing

```
web-07 sshd[2214]: Failed password for invalid user admin from 203.0.113.45 port 52114 ssh2
web-07 sshd[2214]: Accepted publickey for deploy from 198.51.100.20 port 40022 ssh2
web-07 sudo: deploy : TTY=pts/0 ; PWD=/home/deploy ; USER=root ; COMMAND=/usr/bin/systemctl restart nginx
```

> `TargetUsername = admin` `SrcIpAddr = 203.0.113.45` `EventType = Logon` `EventResult = Failure` `DvcHostname = web-07`

1 Find the templates

```
Syslog
| where TimeGenerated > ago(7d)
| where ProcessName == "sshd"
| reduce by SyslogMessage
    with threshold = 0.1
| top 20 by Count
```

Run per program. Test reduce on your lake; Drain3 in a notebook also works.

2 Record each template once

```
- process: sshd
  match: "Failed password for"
  fields:
    TargetUsername: 'for (?:invalid user )?(\S+) from'
    SrcIpAddr: 'from (\S+) port'
  EventType: Logon
  EventResult: Failure
```

One registry entry per template, kept in Git with the generator.

3 Generator writes plain KQL for each template

```
Syslog
| where TimeGenerated > ago(1h)
| where ProcessName == "sshd" and SyslogMessage has "Failed password for"
| extend TargetUsername = extract(@"for (?:invalid user )?(\S+) from", 1, SyslogMessage),
         SrcIpAddr = extract(@"from (\S+) port", 1, SyslogMessage)
| project TimeGenerated, SourceTable = "Syslog", EventVendor = "OpenSSH",
          EventType = "Logon", EventResult = "Failure", SrcIpAddr,
          TargetUsername, DvcHostname = Computer
```

One block per template, unioned into one KQL job. No functions, so it runs in lake KQL. Output columns use ASIM names.

4 Catch every unmatched line

```
Syslog
| where TimeGenerated > ago(1h)
| where not(SyslogMessage has_any
    ("Failed password for", "Accepted"))
| extend Ip = extract_all(
    @"\b(\d{1,3}(?:\.\d{1,3}){3})\b", SyslogMessage)
| mv-expand SrcIpAddr = Ip to typeof(string)
```

Unmatched lines still give up their IPs, as Tier 2 rows.

5 Hunt the index, not the raw table

```
HuntIndex_KQL_CL
| where SrcIpAddr == "203.0.113.45"
| summarize Events = count(),
    Sources = make_set(SourceTable)
    by EventType, EventResult
```

The KQL job writes HuntIndex_KQL_CL on a schedule. Analysts and the data lake MCP query this one table.

Sources: Microsoft Learn, KQL reduce, extract, extract_all and data lake KQL jobs; LevelBlue design

Checked 16 October 2026

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